

ROCKINGHAM COUNTY SHILOH AP, NC, USA

WMO: 742077

Lat: 36.437N

Lon: 79.851W

Elev: 693

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03763

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	20.0	-8.2	3.6	26.7	-0.7	5.4	28.3	16.5	50.1	14.9	49.0	1.1	250	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	93.2	74.5	91.0	73.7	89.2	73.2	78.0	88.1	76.8	86.5	75.6	84.5	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	135.7	82.1	74.1	130.6	80.6	73.1	126.1	79.4	41.8	88.7	40.6	86.5	39.5	84.7	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	11.7	10.1	5.6	97.3	7.8	2.7	0.0	99.2	-4.5	100.8	-8.9	102.3	-14.5	104.3		
			4.8	79.5	7.4	1.6	-0.6	80.7	-4.9	81.7	-9.1	82.6	-14.5	83.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	37.9	41.3	48.7	58.4	67.1	74.1	77.0	75.5	69.8	58.5	47.4	42.5
	DBStd	15.76	9.77	9.79	9.84	8.41	7.05	5.08	4.29	4.77	5.89	7.98	8.30	8.43
	HDD50	1265	392	275	148	23	0	0	0	0	0	18	145	264
	HDD65	3801	841	664	509	229	67	3	0	1	26	232	530	699
	CDD50	4283	17	32	109	274	529	724	837	790	593	281	66	31
	CDD65	1343	0	1	4	30	131	277	372	327	169	30	1	1
	CDH74	12421	0	6	82	447	1234	2671	3456	2853	1356	294	17	5
	CDH80	4872	0	0	6	110	387	1090	1549	1184	490	55	1	0
Wind	WSAvg	2.8	3.4	4.0	4.0	4.1	2.8	2.2	2.0	1.7	1.9	2.4	2.6	2.9
Precipitation	PrecAvg	44.5	3.6	3.0	4.1	3.7	3.6	4.1	4.3	4.2	4.4	3.2	3.0	3.3
	PrecMax	62.6	7.2	5.4	8.9	6.4	7.9	12.2	7.7	8.6	12.2	11.2	8.7	6.4
	PrecMin	31.4	0.8	1.1	0.8	0.6	0.6	0.8	1.7	0.9	0.6	0.0	0.2	0.9
	PrecStd	6.7	1.5	1.1	1.8	1.6	1.9	2.4	1.4	1.7	2.8	2.4	2.0	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.7	74.0	80.8	88.3	90.9	95.3	97.3	96.5	91.4	85.5	76.8	72.7
		MCWB	57.8	60.8	60.1	63.3	70.9	74.8	75.4	71.7	70.2	69.2	57.8	60.2
	2%	DB	63.9	68.4	76.4	83.1	87.3	91.5	93.4	92.4	89.0	81.1	70.8	66.1
		MCWB	54.8	56.4	59.1	63.7	70.1	74.5	75.1	74.1	70.9	65.6	58.7	58.9
	5%	DB	59.2	63.9	71.7	79.5	84.4	89.8	91.0	89.8	86.1	77.2	67.4	61.9
		MCWB	49.9	52.1	56.5	62.0	69.6	73.8	74.9	73.7	71.0	64.7	57.3	55.4
	10%	DB	54.1	58.6	66.8	75.3	81.6	86.6	88.5	86.8	82.3	73.1	63.3	57.2
		MCWB	47.5	47.7	54.5	60.5	68.3	72.7	74.2	73.1	69.6	64.1	54.7	50.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.0	64.1	64.8	69.9	75.0	78.5	79.7	79.0	76.3	73.0	66.5	65.4
		MCDB	65.3	71.4	74.6	78.1	84.8	89.6	90.0	88.7	84.6	77.5	70.0	69.5
	2%	WB	57.2	60.6	62.1	67.2	73.0	76.9	78.2	77.5	74.8	70.7	62.9	61.1
		MCDB	62.0	65.1	70.4	75.8	82.2	88.1	88.7	86.4	83.3	75.8	67.3	63.6
	5%	WB	52.4	55.2	59.8	64.9	71.6	75.6	77.1	76.2	73.3	68.5	59.4	56.2
		MCDB	56.6	61.8	68.1	74.4	80.4	85.9	87.2	84.1	80.9	74.1	64.5	60.5
	10%	WB	47.7	49.5	57.2	63.0	70.2	74.2	75.9	74.9	71.8	65.5	55.7	51.6
		MCDB	53.4	55.3	63.5	72.0	78.3	83.6	85.1	82.1	77.8	71.6	61.3	55.8
Mean Daily Temperature Range	5% DB	MDBR	20.8	21.7	24.1	25.7	23.2	23.0	21.5	20.8	20.8	24.4	24.3	20.2
		MCDBR	27.2	28.0	29.9	30.8	26.9	25.0	24.6	24.5	26.4	27.9	29.3	25.2
		MCWBR	18.3	17.4	15.1	13.3	11.9	10.1	8.9	8.8	10.9	14.5	17.3	17.2
	5% WB	MCDBR	22.2	24.3	23.8	24.1	22.3	23.1	21.9	20.4	21.8	20.6	21.8	20.1
		MCWBR	18.3	18.3	13.5	12.3	10.6	10.2	9.1	8.7	10.2	11.8	15.6	16.3
Clear-Sky Solar Irradiance	taub	0.304	0.320	0.349	0.395	0.435	0.478	0.516	0.513	0.410	0.365	0.320	0.307	
	taud	2.512	2.460	2.425	2.302	2.214	2.132	2.051	2.039	2.325	2.434	2.517	2.538	
	Ebn at Noon	284	292	292	281	271	258	247	245	267	271	275	276	
	Edh at Noon	25	30	34	41	45	49	53	53	37	31	25	23	
All-Sky Solar Radiation	RadAvg	791	1004	1335	1669	1843	1992	1853	1695	1407	1150	872	674	
	RadStd	53	121	89	134	152	147	123	88	133	142	69	59	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (57 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254	+166	

Nomenclature: See separate page